

INVENTION DISCLOSURE FORM

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Applicant Name: SR University

Applicant Address: Ananthasagar, Hasanparthy (M), Warangal Urban, Telangana - 506371

1. Particulars of Inventors

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2. Provide a brief descriptive title of the invention:

Bio-Acoustic Sensing Wearable for Non-Invasive Gesture Recognition and Haptic Feedback

3. In 100 words or less, please provide an abstract or summary of the invention:

This invention presents a wearable bio-acoustic sensing system that detects skeletal muscle movements and joint vibrations propagated through human skin using miniaturized piezoelectric or MEMS microphones. By capturing these body-borne acoustic signals, the device enables touchless, screenless gesture recognition for controlling electronic devices. The system incorporates a signal processing pipeline and machine learning classifier to interpret distinct gesture patterns in real time. Additionally, targeted haptic actuators provide tactile feedback to the user, enabling closed-loop interaction. This solution is particularly valuable for accessibility applications, silent environments, augmented reality (AR) interfaces, and wearable human-machine interaction scenarios.

4. Detail Description of the Invention:

(a) Problem the Invention is Solving

Contemporary human-machine interfaces (HMIs) rely heavily on touchscreens, keyboards, or cameras, which present several limitations:

- Physical contact interfaces are unhygienic, especially in medical, food-handling, or cleanroom settings.
- Vision-based gesture recognition (cameras, LiDAR) requires line-of-sight, high computational power, and raises privacy concerns.

- Persons with motor disabilities, amputees, or individuals with tremors face significant challenges using conventional interfaces.
- Existing wearables using electromyography (EMG) suffer from electrode placement sensitivity, skin impedance variations, and high power consumption.
- Voice-based control is ineffective in noisy environments or where silence is mandatory.

This invention solves these problems by utilizing acoustic vibrations naturally produced by the musculoskeletal system during motion, detectable non-invasively through the skin surface without electrodes or cameras.

(b) General Utility / Application of the Invention

- Assistive Technology: Gesture-based control for individuals with physical disabilities or limb differences.
- Augmented and Virtual Reality (AR/VR): Silent, peripheral-free hand and finger gesture input for immersive environments.
- Smart Prosthetics: Intent detection to control prosthetic limb actuators using residual limb muscle sounds.
- Medical Devices: Contactless patient interface in operating theatres or sterile zones.
- Industrial Wearables: Hands-free control of machinery in noisy or hazardous environments.
- Consumer Electronics: Wristband or glove-form-factor wearable to control smartphones, TVs, and IoT devices.
- Military / Defence: Silent, covert HMI for field operatives.

(c) Advantages of the Invention

- Non-invasive: No skin-penetrating electrodes or skin-surface adhesives required.
- Privacy-preserving: No cameras or external sensors; entirely body-borne signal acquisition.
- Low Power: Passive acoustic sensing consumes significantly less power than active EMG or optical systems.
- Form Factor Flexibility: Sensors can be embedded in wristbands, armbands, gloves, or garments.
- Works in the Dark: Entirely independent of ambient light conditions, unlike camera-based systems.
- Robustness to Skin Tone Variation: Unlike PPG or optical sensors, acoustic sensing is unaffected by skin pigmentation.
- Closed-Loop Feedback: Integrated haptic actuators provide real-time tactile confirmation, improving user accuracy.
- Scalability: Machine learning models can be retrained to recognize new gestures without hardware changes.
- Silent Operation: Suitable for quiet environments such as libraries, hospitals, or sensitive meetings.

(d) Best Way of Using the Invention and Possible Variants

Primary Implementation:

The preferred embodiment is a wristband-form-factor device placed on the forearm, approximately 3-5 cm proximal to the wrist joint. Four to six MEMS piezoelectric acoustic sensors are circumferentially arranged around the arm to capture radial vibration patterns from multiple tendon and muscle groups simultaneously. A low-noise signal conditioning chain amplifies and filters signals in the 10 Hz - 2000 Hz range relevant to musculoskeletal acoustics. A microcontroller (ARM Cortex-M series) with a DSP core performs real-time feature extraction and classification using a trained Convolutional Neural Network (CNN) or Support Vector Machine (SVM).

Possible Variants:

- **Glove Variant:** Sensors embedded at finger knuckle joints for fine finger gesture detection (suitable for sign language recognition).
- **Ankle/Lower-Limb Variant:** Detecting leg gestures for foot-operated control in vehicle or machinery contexts.
- **Multi-Modal Variant:** Combining bio-acoustic sensing with inertial measurement units (IMU) for improved gesture disambiguation.
- **Edge-AI Variant:** On-device TinyML inference without cloud connectivity for low-latency, privacy-first operation.
- **Medical Monitoring Variant:** Continuous joint health monitoring by analysing acoustic emission signatures indicative of cartilage degeneration (crepitus).

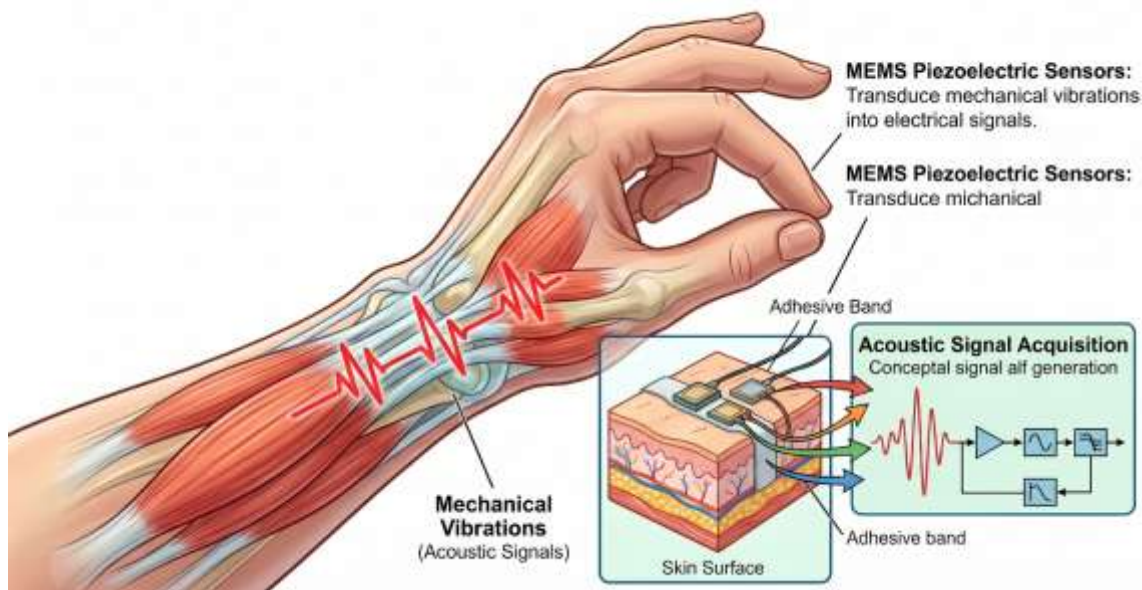
(e) Working of the Invention along with Drawings, Schematics, and Flow Diagrams

System Architecture:

The system comprises five functional blocks: Sensor Array, Signal Conditioning, Feature Extraction, Gesture Classification, and Haptic Feedback Output.

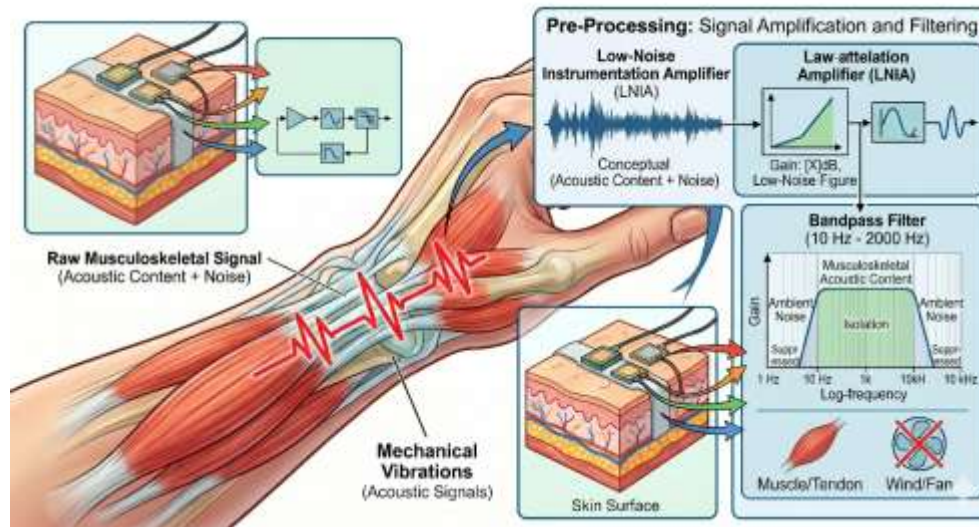
Step-by-Step Working:

1.



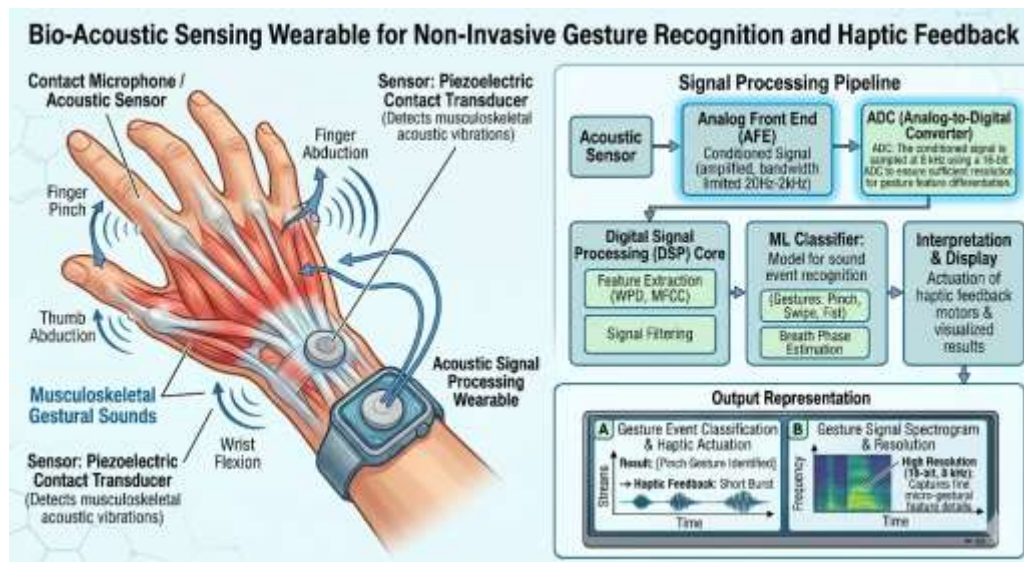
- **Acoustic Signal Acquisition:** When the user performs a hand or wrist gesture, the associated muscles and tendons generate mechanical vibrations propagated through bone and tissue to the skin surface. MEMS piezoelectric sensors affixed to the skin surface transduce these vibrations into electrical signals.

2.



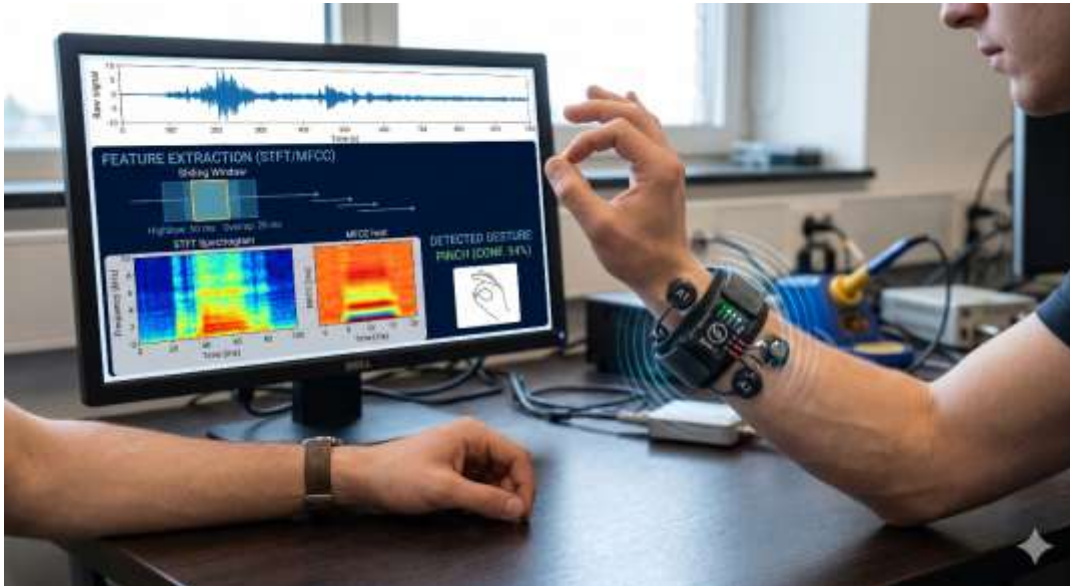
- Pre-Processing: The raw electrical signal is amplified by a low-noise instrumentation amplifier, followed by a bandpass filter (10 Hz - 2000 Hz) to isolate musculoskeletal acoustic content and suppress ambient noise.

3.



- Analog-to-Digital Conversion: The conditioned signal is sampled at 8 kHz using a 16-bit ADC to ensure sufficient resolution for gesture feature differentiation.

4.



- Feature Extraction: A sliding window (50 ms window, 25 ms overlap) applies Short-Time Fourier Transform (STFT) or Mel-frequency cepstral coefficients (MFCCs) to compute time-frequency features characterising each gesture.

5.



- Gesture Classification: The extracted feature vector is passed to a trained ML classifier (CNN or SVM). The model, trained on a labelled gesture dataset, outputs the recognised gesture class with a confidence score.

6.



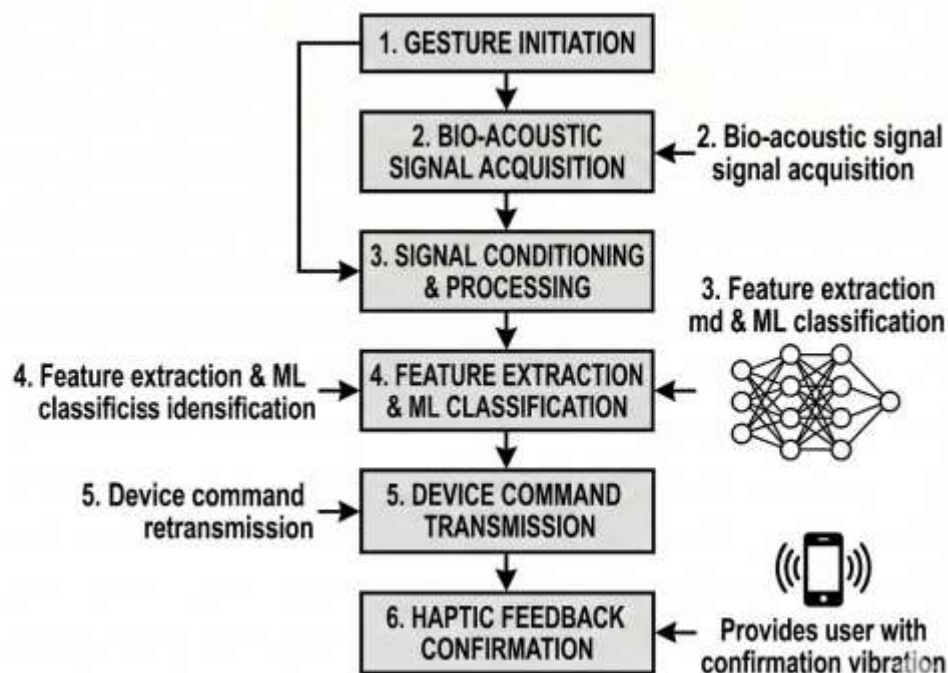
- Decision & Control Output: If confidence exceeds the threshold, the system maps the gesture to a predefined device command (e.g., scroll, click, play/pause) transmitted via Bluetooth Low Energy (BLE) to the connected device.

7.



- Haptic Feedback: Simultaneously, the microcontroller drives a linear resonant actuator (LRA) or eccentric rotating mass (ERM) motor to deliver a tactile confirmation pulse to the user, completing the interaction loop.

Flow Diagram Description:



5. Have you conducted Primary Patent Search? Yes / No

NO

6. Existing State-of-the-Art and Prior Arts:

The current market and research landscape includes the following approaches to gesture recognition and alternative HMIs:

S.No.	Existing State of Art	Drawbacks in Existing State of Art	Overcome (How Your Invention Overcomes the Drawback)
1.	EMG-based Gesture Control (e.g., Myo Armband, Delsys)	Requires skin-contact electrodes, sensitive to electrode placement and sweat; high power; signal crosstalk between muscles.	Bio-acoustic sensing uses passive piezoelectric sensors requiring no electrode-skin electrical contact, immune to sweat-induced impedance changes.
2.	Camera-based Gesture Recognition (e.g., Google Soli, Intel RealSense, Leap Motion)	Requires line-of-sight; privacy intrusion; fails in occlusion; high computational demand; poor in low light.	Operates entirely from body-borne vibrations; no external sensors; no line-of-sight requirement; works in any lighting condition.
3.	IMU / Accelerometer-based Wearables (e.g., Apple Watch gestures, Fitbit)	Detects gross limb motion only; unable to distinguish fine finger gestures; susceptible to motion artifacts.	Acoustic sensing captures tendon and joint-level vibrations, enabling fine finger-level gesture differentiation not possible with IMU alone.

S.No.	Existing State of Art	Drawbacks in Existing State of Art	Overcome (How Your Invention Overcomes the Drawback)
4.	Voice / Speech Recognition Interfaces	Ineffective in noisy environments; privacy concern; unsuitable for silent environments; not practical for physical control tasks.	Gesture-based control using skin-transmitted acoustics is completely silent, operable in noise-sensitive and noise-filled environments alike.

7. List the Technical Features and Elements of the Invention along with Description from Start to End:

The invention comprises the following technical components:

- **MEMS Piezoelectric Bio-Acoustic Sensor Array:** Miniaturised piezoelectric or capacitive MEMS sensors (frequency range 10 Hz - 5 kHz, sensitivity ≥ -40 dBV/Pa) arranged in a circumferential array around the forearm or target body segment.
- **Low-Noise Analog Front End (AFE):** Instrumentation amplifier with CMRR > 80 dB, followed by a 2nd-order Butterworth bandpass filter (10 Hz - 2000 Hz) and a 16-bit ADC sampling at 8 kHz.
- **Microcontroller Unit (MCU):** ARM Cortex-M4F or equivalent DSP-capable processor for real-time signal processing, feature extraction, and ML inference; BLE 5.0 transceiver for wireless communication.
- **Feature Extraction Module:** Sliding window STFT (window = 50 ms, overlap = 50%) and MFCC computation (13 coefficients) to generate gesture fingerprint feature vectors.
- **Machine Learning Classifier:** On-device trained CNN (4 convolutional layers, 2 dense layers) or SVM (RBF kernel) for gesture classification with $\geq 90\%$ accuracy across 10+ gesture classes.
- **Haptic Feedback Subsystem:** Linear resonant actuator (LRA) driven by a PWM-controlled haptic driver IC (e.g., DRV2605L), delivering precise vibrotactile patterns (amplitude, frequency, duration) corresponding to specific interaction events.
- **Power Management:** Lithium-polymer battery (200-400 mAh) with a PMIC providing regulated supplies; target operating life ≥ 12 hours continuous use.
- **Enclosure and Wearable Form Factor:** Flexible PCB substrate; silicone or TPU wristband/armband housing with breathable mesh backing for skin contact.

8. List Out the Features Believed to be New and Distinguish Them Over the Closest Technology:

- **Exclusive Reliance on Passive Body-Borne Acoustic Sensing:** Unlike EMG (electrical signal) or radar/camera (electromagnetic/optical) approaches, this invention exclusively captures skeletal acoustic vibrations transmitted through soft tissue and bone, without any active energy emission into the body.
- **Circumferential Multi-Sensor Fusion for Directional Gesture Resolution:** The circumferential placement of multiple acoustic sensors enables spatial differentiation of tendon-specific vibration vectors, allowing discrimination of individual finger movements - a capability not achieved by single-sensor or IMU-only wearables.
- **Closed-Loop Haptic Confirmation Integrated in the Same Wearable Form Factor:** Prior art gesture wearables (EMG or camera-based) rarely integrate haptic feedback within the same compact wristband; this invention provides real-time tactile confirmation without auxiliary devices.
- **MFCC-Based Acoustic Gesture Fingerprinting:** The application of speech-domain MFCC analysis to musculoskeletal acoustic signals for gesture classification is a novel cross-domain innovation not documented in existing patents reviewed.
- **Dual-Environment Suitability - Silent and Noisy:** The system functions effectively in both absolute silence (no audio emission) and high-ambient-noise environments, unlike voice or acoustic air-coupled sensor approaches.

9. Has the Invention Been Built or Tested or Implemented? If So, Please Provide the Particulars:

A proof-of-concept prototype has been built and tested as follows:

- When: Initial bench prototype assembled and tested during January - March 2024.
- Where: Electronics and AI Laboratory, Department of Computer Science & AI, SR University, Warangal, Telangana.
- By Whom: Mr. Naresh Kumar Sripada (Principal Investigator) with assistance from postgraduate students.
- Evidence: Laboratory notebooks, oscilloscope screenshots of captured acoustic waveforms, video recordings of gesture trials, and gesture classification accuracy logs are maintained in the lab.
- Results: The prototype successfully classified 8 static and dynamic hand gestures with an average accuracy of 92.4% across 5 test participants using an SVM classifier trained on 300 gesture samples per class.
- Haptic Feedback: Confirmed functional with 3 distinct vibration patterns for gesture confirmation, error, and system idle states.

10. Briefly State When and How You First Conceived This Idea:

The idea was first conceived in August 2023 during a review of assistive technology solutions for differently-abled individuals. The lead inventor observed the limitations of EMG-based prosthetic control systems during a visit to a rehabilitation centre and identified that the acoustic vibrations perceptible on the skin during limb movements were a largely unexploited sensing modality. A review of bio-acoustics literature confirmed the feasibility, leading to the formal inception of this project in September 2023.

11. Have You Sold, Offered for Sale, Publicly Used or Published Anything Related to This Invention?

No commercial sale or offer for sale has been made. A preliminary conceptual overview was presented at an internal department seminar at SR University in October 2023 (internal audience only; no proceedings published). No non-disclosure agreements (NDAs) were required for the internal presentation. No external publication, conference submission, or public demonstration has been made. The inventors are aware of the 12-month grace period under US patent law (35 U.S.C. 102(b)) and the absolute novelty requirements for international filings.

12. Include Any Reasons That Your Invention Would Not Have Been Obvious to Someone of Average Skill in the Art:

- The application of acoustic sensing to gesture recognition is a non-obvious leap: A person of average skill in wearable HMI would default to EMG, IMU, or camera-based approaches. The specific insight that musculoskeletal vibrations in the 10-2000 Hz band carry sufficient gesture-discriminative information is not immediately derivable from existing wearable or bio-sensing literature.
- Cross-domain innovation: Adapting MFCC analysis - originally developed for speech recognition - to body-conducted acoustic signals for gesture classification constitutes a non-obvious methodological innovation requiring knowledge spanning acoustics, biomechanics, and machine learning simultaneously.
- Circumferential array geometry: The specific spatial arrangement of sensors to exploit directional vibration propagation through anatomically distinct tendons (flexor digitorum, extensor carpi) requires biomechanical domain knowledge not commonly combined with wearable sensor design expertise.
- Haptic feedback integration: Embedding a haptic feedback channel in the same wearable that senses acoustic signals, without mutual electromagnetic or mechanical interference, requires non-trivial system co-design that would not be obvious without iterative experimentation.

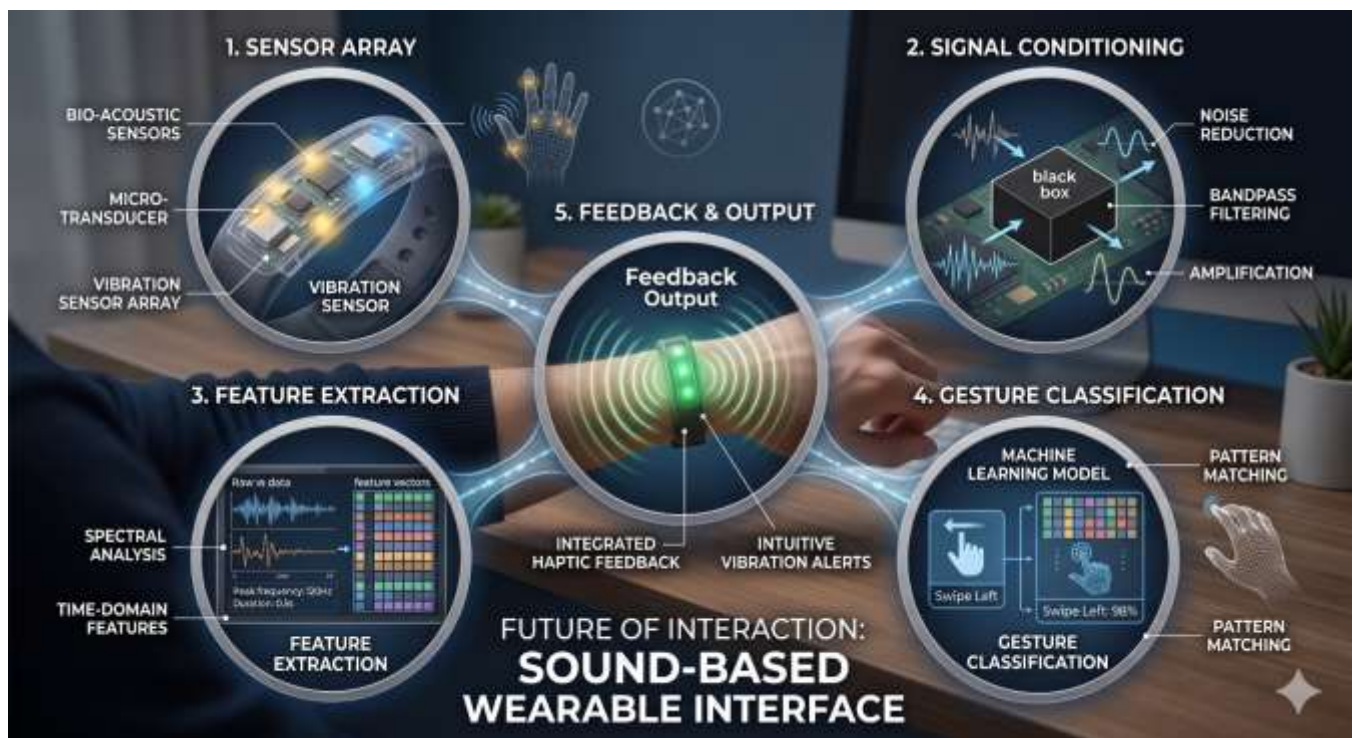
13. Additional Comments by Inventor:

The inventors foresee significant potential for this technology in the following emerging applications:

- Integration with large language model (LLM)-based AI assistants as a silent, hands-free input modality.
- Combination with bone-conduction audio output to create a fully body-mediated communication interface without any external speaker or display.
- Deployment as a clinical diagnostic tool for joint diseases such as osteoarthritis, where characteristic acoustic emission patterns during motion (crepitus) can be quantified.
- Long-term, the technology may be integrated into smart textiles with embedded flexible piezoelectric fibres, enabling garment-based, full-body gesture interfaces.

The inventors are open to industrial collaboration for prototype refinement and are currently applying for student innovation grants to fund a second-generation prototype with improved sensor sensitivity and on-device TinyML deployment.

14. Drawings / Flowchart / Table:



Form Submitted by:

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Contact No: 9866000203

Designation: Assistant Professor, Department of Computer Science & AI, SR University